R&S AUCTIONS

Online Auction Management System

Mini Project Documentation

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1. Abstract

R&S Auctions is a web-based online auction platform developed to facilitate secure and efficient buying and selling of products through an auction mechanism. The system allows users to register as bidders, sellers, or both. Sellers can create auctions by uploading product details and images, while bidders can place bids on active auctions. The platform automatically updates the current highest bid and determines the winner based on the highest bid placed. The system provides role-based access, auction management, bid tracking, winner declaration, and an intuitive user interface for seamless interaction.

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2. Introduction

Online auctions have become a popular method for buying and selling products over the internet. Traditional auction systems require physical presence and manual management, which can be time-consuming and inefficient. The R&S Auctions platform addresses these challenges by providing a digital solution that enables users to participate in auctions from anywhere.

The system allows sellers to create auctions and bidders to compete for products by placing bids. It provides transparency, accessibility, and convenience while maintaining a secure environment for all participants.

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3. Problem Statement

Traditional auction systems suffer from several limitations:

- Limited accessibility due to physical location constraints.
- Manual tracking of bids and auction results.
- Lack of transparency in bidding processes.
- Time-consuming auction management.
- Difficulty in handling large numbers of participants.

The proposed system aims to overcome these challenges by providing an online platform that automates auction management and bidding activities.

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4. Objectives

The main objectives of the project are:

- To develop an online auction platform.

- To allow users to register and log in securely.
- To provide role-based access control.
- To enable sellers to create and manage auctions.
- To allow bidders to place bids on active auctions.
- To maintain auction records and bidding history.
- To determine auction winners automatically.
- To provide an interactive and user-friendly interface.

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5. Scope of the Project

The scope of the project includes:

- User Registration and Authentication
- Role-Based Access Management
- Auction Creation and Management
- Bid Placement and Tracking
- Winner Declaration
- Dashboard Analytics
- Admin Management Features

Future enhancements may include:

- Online Payment Integration
- Real-Time Notifications
- Email Alerts
- Chat System
- Mobile Application Support

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6. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4GB minimum
- Storage: 10GB free space
- Internet Connection

Software Requirements

- Operating System: Windows 10/11
- Java JDK 21
- Spring Boot 3
- MySQL Database
- React JS
- Node.js
- Maven
- Visual Studio Code
- Git and GitHub

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7. Technologies Used

Frontend

- React JS
- React Router DOM
- Axios
- CSS3

Backend

- Spring Boot
- Spring Data JPA
- REST APIs
- Maven

Database

- MySQL

Version Control

- Git
- GitHub

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8. System Architecture

The system follows a three-tier architecture:

Presentation Layer

React-based user interface.

Business Logic Layer

Spring Boot REST APIs handling authentication, auctions, bidding, and winner determination.

Data Layer

MySQL database storing users, auctions, and bid information.

Flow:

User → React Frontend → Spring Boot Backend → MySQL Database

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9. Modules

User Management Module

Features:

- User Registration
- User Login
- Role Assignment
- Session Management

Roles:

- Bidder
- Seller
- Both
- Admin

Auction Management Module

Features:

- Create Auctions
- View Auctions
- Manage Auction Details
- Upload Product Images

Bidding Module

Features:

- Place Bids
- View Current Highest Bid

- Bid History Tracking
- Bid Validation

Dashboard Module

Features:

- Total Auctions
- Active Auctions
- Closed Auctions
- Highest Bid Information
- Seller Auction Statistics

Winner Module

Features:

- Determine Highest Bidder
- Display Winner Details
- Display Winning Amount

Admin Module

Features:

- Monitor Auctions
- Remove Incorrect Auctions
- Manage Platform Activities

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10. Database Design

Users Table

Attributes:

- id
- name
- email
- password
- role

- created_at

Auctions Table

Attributes:

- id
- title
- description
- starting_price
- current_price
- start_time
- end_time
- status
- image_url

Bids Table

Attributes:

- id
- auction_id
- user_id
- bid_amount
- bid_time

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11. Implementation

The frontend was developed using React JS with reusable components and routing. The backend was developed using Spring Boot REST APIs and connected to a MySQL database using Spring Data JPA. Users can create accounts, log in, create auctions, place bids, and view auction results.

Role-based functionality ensures that only sellers can create auctions, bidders can participate in bidding, and administrators can manage platform activities.

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12. Testing

The application was tested for:

- User Registration
- User Login
- Auction Creation

- Auction Viewing
- Bid Placement
- Winner Determination
- Dashboard Functionality
- Admin Access Control

All modules were tested successfully.

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13. Results

The developed system successfully provides an online auction environment where sellers can create auctions and bidders can participate competitively. The system accurately tracks bids, updates auction status, and determines winners based on the highest bid amount.

The platform improves efficiency, transparency, and accessibility compared to traditional auction systems.

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14. Conclusion

R&S Auctions successfully demonstrates the implementation of a modern online auction platform using React JS, Spring Boot, and MySQL. The project achieves all intended objectives and provides a secure, efficient, and user-friendly auction environment.

The platform can be further enhanced with real-time notifications, payment gateway integration, and mobile support to provide a complete commercial solution.

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15. References

1. React JS Documentation - https://react.dev
2. Spring Boot Documentation - https://spring.io/projects/spring-boot
3. MySQL Documentation - https://dev.mysql.com/doc
4. Maven Documentation - https://maven.apache.org
5. GitHub Documentation - https://docs.github.com